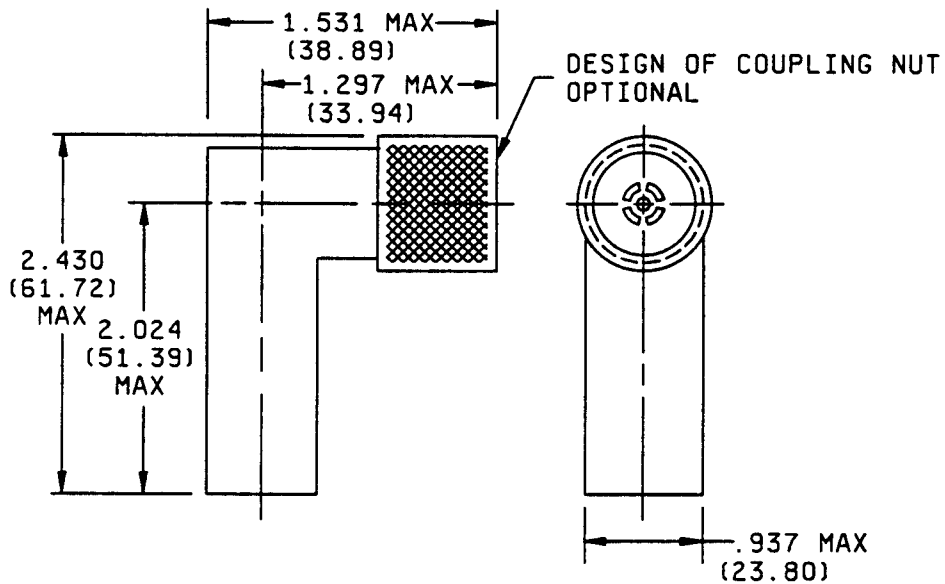
NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time the document is revised.

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY
(SERIES N (CABLED) - RIGHT ANGLE, PIN CONTACT, CLASS 2)
FOR SEMIRIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

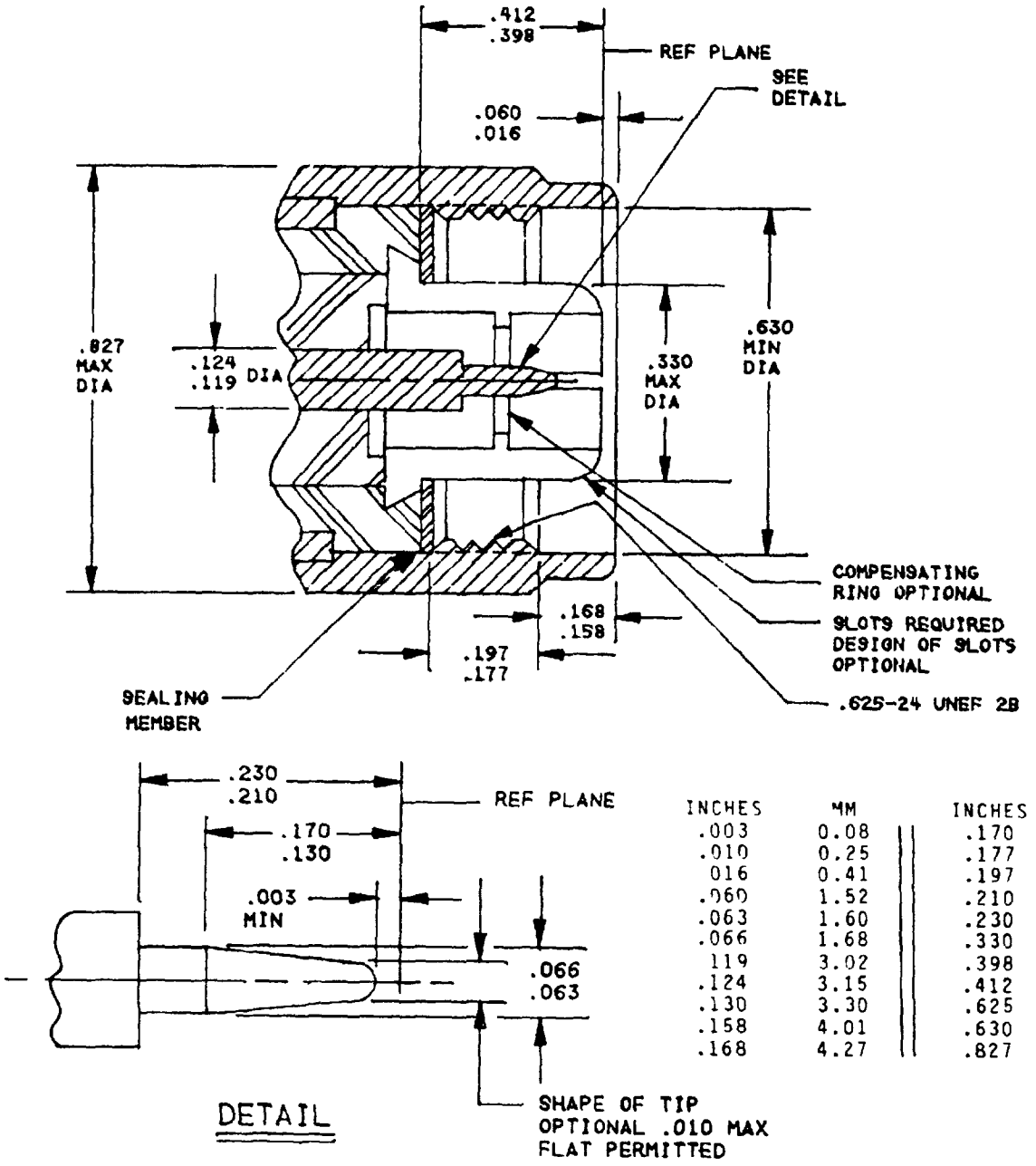
The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-PRF-39012.



NOTES:
1. Dimensions are in inches.
2. Dimension 1.531 (38.89 mm), .937 (23.80 mm) and 2.430 (61.72 mm) are the largest overall dimensions of the connector.
3. Wrench flats, when applicable, are to accommodate standard wrench opening per FED-STD-H28.
4. All undimensioned pictorial representations are for reference purposes only.
5. Dimension 2.430 (61.72 mm) defines the maximum length of the connector when assembled to the appropriate cable.
6. Metric equivalents are given for general information only.
7. Metric equivalents are in parentheses.

FIGURE 1. General configuration.

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INCHES	MM	INCHES	MM
.003	0.08	.170	4.31
.010	0.25	.177	4.50
016	0.41	.197	5.00
.060	1.52	.210	5.33
.063	1.60	.230	5.84
.066	1.68	.330	8.38
119	3.02	.398	10.11
.124	3.15	.412	10.46
.130	3.30	.625	15.88
.158	4.01	.630	16.00
.168	4.27	.827	21.01

NOTES
1. Dimensions are in inches.
2. Three holes .027 (0.69 mm) minimum diameter, equally spaced, are required for safety wiring after mating. Location on coupling nut optional.
3. Metric equivalents are given for general information only.

FIGURE 2. Mating dimensions for pin terminations.

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TABLE I. Dash number and applicable cable.

Dash No.	Applicable cable 2/
CATEGORY A - FIELD SERVICEABLE 1/ (NO SPECIAL TOOLS REQUIRED)	
0001 3001	M17/130-RG402* 0.141 dia. (3.58 mm) M17/130-00001, 00002, 00003
0002 3002	M17/129-RG401* 0.250 dia. (6.35 mm)
CATEGORY E - FIELD REPLACEABLE 1/ (STANDARD ASSEMBLY TOOL KIT)	
0003 3003	M17/130-RG402* 0.141 dia. (3.58 mm) M17/130-00001, 00002, 00003

Dash No.	Applicable cable 2/	Tool No.	Positioning dies	Locator pins	Crimp dies M22520/5
CATEGORY F - FIELD REPLACEABLE MIL-C-22520 (COMPRESSION TOOL) 1/					
3201	M17/129-RG401*	M22520/ 56-01	M22520/36-XX	M22520/36-XX	---
3202	M17/130-RG402* M17/130-00001, 00002, 00003				
3203	M17/129-RG401*	M22520/ 5-01	--	---	-XX
3204	M17/130-RG402* M17/130-00001, 00002, 00003				

1/ These connectors have captivated center contacts.
2/ The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable.

NOTE Direct solder type connectors shall not be used on cable larger than .141 nominal diameter.

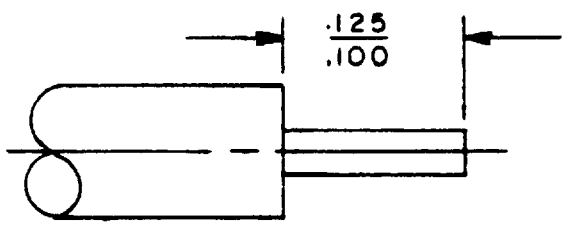


FIGURE 3 Cable stripping dimensions for category E and category F connectors.

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ENGINEERING DATA

Nominal impedance 50 ohms

Frequency range 0 to 18 GHz.

Voltage rating

500 volts rms maximum working voltage at sea level.

125 volts rms maximum at 70,000 feet.

Temperature rating

-65°C to +105°C for M17/129-RG401.

-65°C to +105°C for M17/130-RG402, M17/130-00001, 00002, 00003.

REQUIREMENTS

Dimensions and configuration See figure 1.

Force to engage and disengage

Longitudinal force Not applicable.

Torque: 6 inch-pounds maximum.

Coupling proof torque 15 inch-pounds minimum.

Inspection conditions. For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 6 to 10 inch-pounds.

Mating characteristics See figure 2 for dimensions.

Outer contact

Test ring I. D.: .316 maximum, 16 microinches finish

Insertion force 25 pounds maximum when inserted a minimum of .093.

Contacts with slotted members Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Insulation resistance Method 3003 of MIL-STD-1344; 5,000 megohms minimum.

Center contact retention (applicable to captivated center contact connectors only) 6 pounds minimum axial force.

Corrosion (salt spray) Method 1001 of MIL-STD-1344, test condition B.

Voltage standing wave ratio (VSWR) From 2.0 to 18.0 GHz, 1.15 +.015 freq. (GHz).

Swept frequency VSWR test setup

Item 6: VSWR shall be less than 1.008 +.002 F (F in GHz).

Item 16. VSWR shall be less than 1.008 +.002 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.012 +.004 F (F in GHz).

Group B inspection VSWR shall be less than 1.024 +.007 F (F in GHz)

Qualification and group C inspection VSWR shall not exceed 1.08.

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Connector durability: Method 2016 of MIL-STD-1344.

500 cycles at 12 cycles per minute maximum.

The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance (in milliohms maximum).

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable
Outer contact (stainless steel)	2.0	Not applicable
Cable outer conductor to body	.05	Not applicable
Cable outer conductor to stainless steel body	2.0	Not apolicable

Dielectric withstanding voltage. Method 3001 of MIL-STD-1344; 1,500 volts rms minimum at sea level.

Vibration, high frequency. Method 2005 of MIL-STD-1344, test condition IV, except the method of mounting shall be approved by the qualifying activity, and 30 g peak will be applicable.

Shock: Method 2004 of MIL-STD-1344, test condition G.

Thermal shock: Method 1003 of MIL-STD-1344, test condition A, except test high temperature shall be +115°C.

Humidity Method 1002 of MIL-STD 1344. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude 70,000 feet.

Cable (semirigid)	V rms
M17/129-RG401	500
M17/130-RG402, M17/130-00001, 00002, 00003	375

RF high potential withstanding voltage.

Frequency: 5 to 7.5 MHz inclusive.

Leakage current Not applicable.

Cable (semirigid)	V rms
M17/129-RG401	1,200
M17/130-RG402, M17/130-00001, 00002, 00003	1,000

Cable retention

Cable	Force	Torque
M17/129-RG401	90 pounds minimum	70 inch-ounces minimum
M17/130-RG402, M17/130-00001, 00002, 00003	60 pounds minimum	55 inch-ounces minimum

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Coupling mechanism retention force: 100 pounds minimum.

RF leakage -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss dB maximum = 15 X $\sqrt{\text{freq. GHz.}}$ Test frequency at 6.0 GHz.

Part number M39012/130- (dash number from table I).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following dash numbers 1/ 2/	Qualifies the following dash numbers
I	*001	*001
II	*002	*002
III	*003	*003

1/ Individual connectors other than listed are self qualifying only.

2/ Qualification of connectors qualifies connectors of the same material only.

* Body material and connector requirement.

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - MI, FA
Navy - SH
Air Force - 11, 17, 80, 99
DLA - ES

User activities
Army - AT, AV
Navy - AS, MC, OS
Air Force - 19

Agent
DLA - ES

Preparing activity
Army - CR

(Project 5935-3438-02)

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